

CASE STUDY: SUPER TYPHOON ROLLY (GONI), 2020

Course: METEOR 2110 – Introduction to Meteorology

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I. Introduction

Super Typhoon Rolly, internationally known as Goni, was one of the most intense tropical cyclones ever recorded in the Western Pacific. On November 1, 2020, it made landfall in the Philippines with catastrophic winds and rainfall, marking the strongest landfalling cyclone globally for that year. The storm's rapid intensification and destructive power offer valuable insight into the behavior of tropical systems under near-ideal oceanic and atmospheric conditions. Studying such events is crucial to understanding meteorological dynamics, improving disaster preparedness, and refining forecast accuracy. This report presents a detailed meteorological case study covering Rolly's formation, track, characteristics, impacts, and relevance to core meteorological concepts.

II. Cyclone Identification and General Overview

Typhoon Goni, locally named Rolly, formed over the warm waters east of Micronesia on October 26, 2020. It entered the Philippine Area of Responsibility (PAR) at around 11:00 PM on October 29, 2020, and rapidly intensified into a Super Typhoon. According to PAGASA (2021) and JTWC (2021), Rolly's peak sustained winds reached 315 km/h (1-minute average), with a central pressure of 905 hPa. It made four landfalls: first in Bato, Catanduanes; then in Tiwi, Albay; San Narciso, Quezon; and finally emerging into the West Philippine Sea on November 1 before exiting the PAR on November 3. Figure 1 illustrates the track of Super Typhoon Rolly as monitored by PAGASA and international agencies.

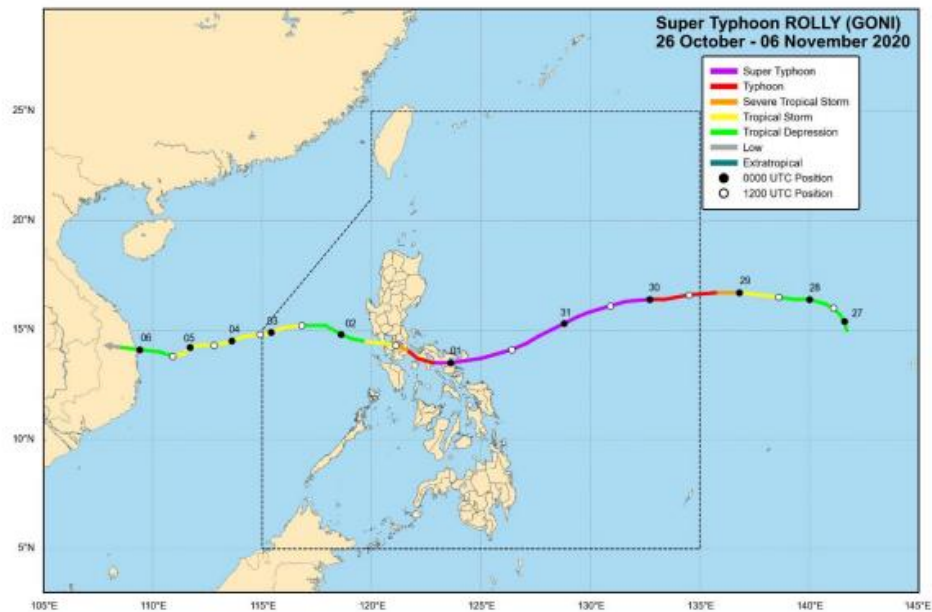


Figure 1. Track of Super Typhoon Rolly (Goni) as it crossed the Philippines (PAGASA, 2021).

III. Meteorological Background and Cyclogenesis

Rolly developed under nearly perfect environmental conditions for cyclogenesis. Sea surface temperatures (SST) in the Philippine Sea reached 30–31 °C, providing an abundant source of latent heat energy. The atmosphere exhibited weak vertical wind shear (<10 knots), allowing deep convection to remain vertically aligned. Upper-level divergence was strong, enhancing outflow and promoting rapid pressure falls at the surface. Figure 2 shows a Himawari-8 infrared satellite image highlighting the storm's tightly wound structure prior to landfall. The Coriolis effect, coupled with high moisture and instability, facilitated rotation and sustained convective development.

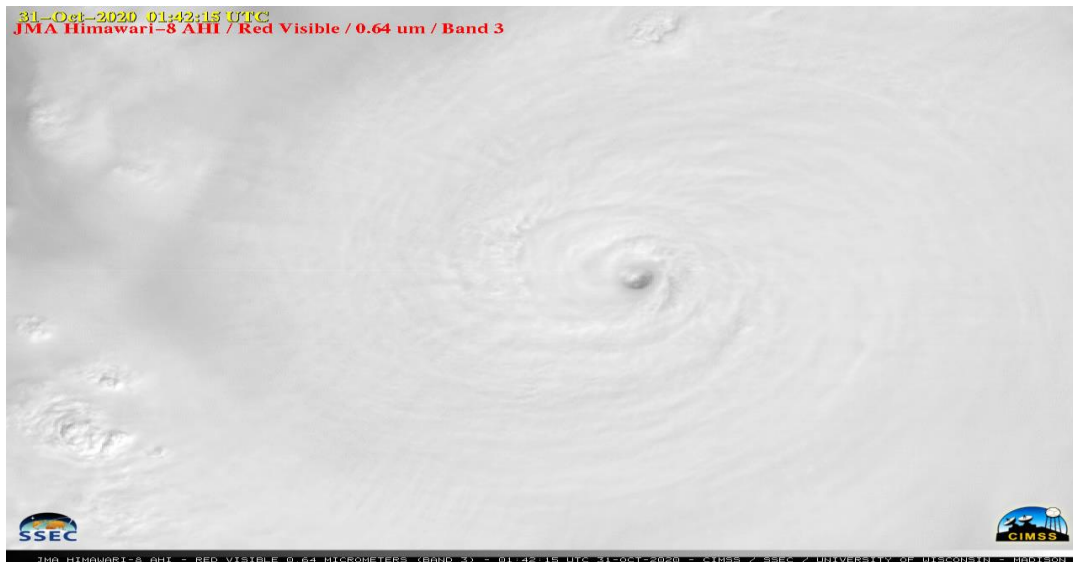


Figure 2. Himawari-8 satellite infrared image showing compact eyewall structure before landfall (CIMSS, 2020).

Sea Surface Temperature (SST) Analysis

The Sea Surface Temperature (SST) field in the western Pacific during late October 2020 was exceptionally favorable for tropical cyclone development. Data from NOAA and JMA indicated that the waters east of the Philippines maintained temperatures of 30–31 °C, extending deep into the mixed layer. This created a high oceanic heat content (OHC) reservoir, sustaining convection even after repeated bursts of latent heat release. Figure 3a shows the spatial SST anomaly distribution, with positive deviations exceeding +1.5 °C above climatological averages. These anomalies coincided with an ongoing La Niña phase, which enhanced trade winds and induced warm water piling in the western Pacific. The thermal structure of the ocean effectively reduced upwelling of cold water beneath the cyclone, maintaining high enthalpy fluxes necessary for rapid intensification. This warm pool not only fueled Rolly's rapid intensification but also supported the subsequent storms that developed in the 2020 typhoon season. In essence, the preconditioning of the ocean was a key contributor to Rolly's remarkable strength. The consistent surface flux of moisture and heat between the ocean and atmosphere reinforced the storm's internal thermodynamics, maintaining its Category 5 status for an extended period.

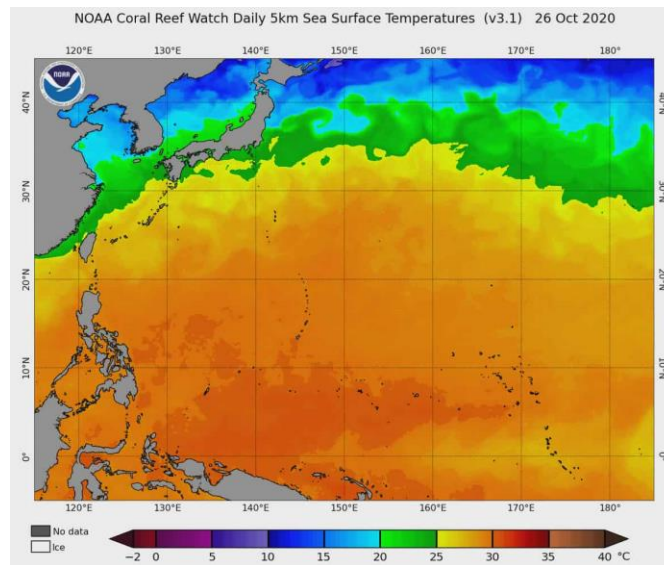


Figure 2a. NOAA SST anomaly map showing warm waters during Rolly's genesis (NOAA, 2020).

Vertical Wind Shear Environment

Vertical wind shear — the change in wind speed and direction with height — plays a critical role in maintaining a tropical cyclone's structure. For Super Typhoon Rolly, the Himawari-8 infrared sequence (*Figure 2b*) clearly illustrates how favorable shear conditions allowed the storm to rapidly intensify within 24 hours. On October 29, 2020, the system appeared disorganized, but by October 30, it had developed a compact, symmetric eyewall as shear values across the Philippine Sea remained below 8 knots. This vertically aligned structure enabled efficient upward motion and strong outflow aloft, sustaining explosive deep convection around the eye. However, by November 1, increased shear and land interaction over Luzon caused the storm's rapid weakening, as seen in the degradation of the eyewall and loss of symmetry. The imagery underscores how even small changes in wind shear can determine whether a cyclone intensifies or collapses.

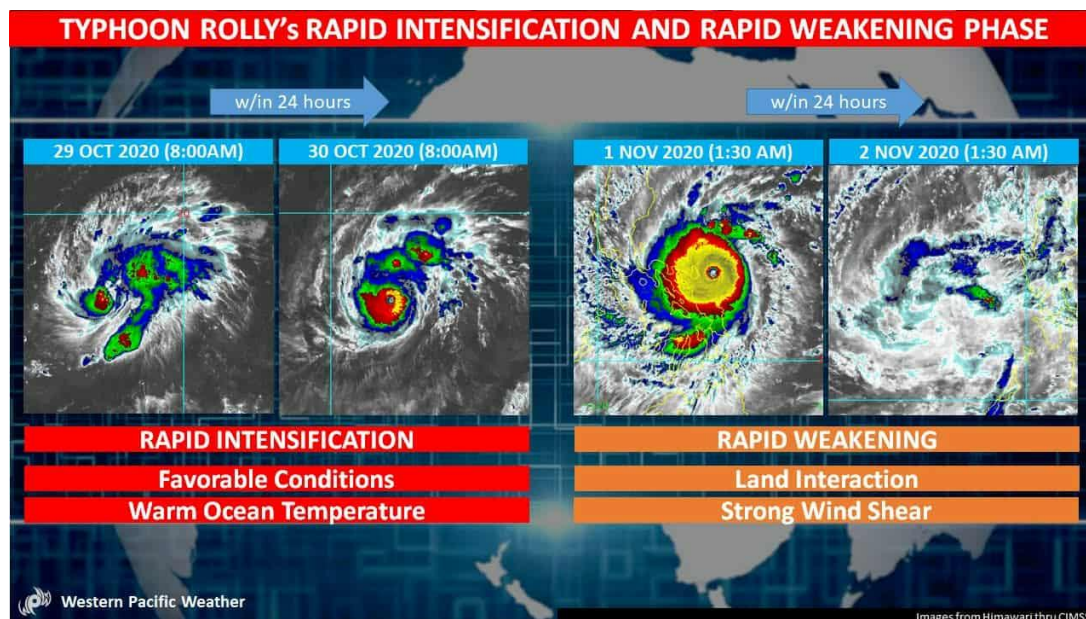


Figure 2b. Himawari-8 infrared sequence showing Super Typhoon Rolly's rapid intensification and weakening phase (29October–2November2020).

Source: Western Pacific Weather using Himawari-8 data via CIMSS (2020).

IV. Cyclone Characteristics and Dynamics

Rolly underwent a remarkable phase of explosive intensification, strengthening by approximately 55 knots within just 24 hours, a textbook example of rapid intensification (RI) driven by ideal atmospheric and oceanic conditions. This rate of intensification far exceeded the standard RI threshold of 30 knots in 24 hours, underscoring the storm's exceptional thermodynamic efficiency. Satellite imagery and Dvorak analysis from the Joint Typhoon Warning Center (JTWC, 2021) revealed a sharply defined, pinhole eye less than 20 km wide, surrounded by a compact eyewall of deep convection with cloud-top temperatures below -80°C , signifying extreme vertical motion and latent heat release. The Japan Meteorological Agency (JMA, 2020) simultaneously reported 10-minute sustained winds of 220 km/h, while JTWC's assessment of 1-minute sustained winds at 315 km/h confirmed Rolly as the strongest landfalling tropical cyclone globally in 2020, equivalent to a Category 5 system on the Saffir–Simpson scale. The cyclone's small radius of maximum winds (RMW), estimated at roughly 15–20 km, highlighted its compact yet ferocious wind field, where the most intense pressure gradient and destructive winds were tightly concentrated near the eye. As illustrated in *Figure 3*, PAGASA's analysis of wind fields and surface pressure gradients showed a symmetrically organized circulation with nearly perfect concentric isobars—a feature indicative of a mature, thermodynamically balanced warm-core system. The maintenance of this intense structure was supported by sustained upper-level divergence and efficient oceanic heat flux, which together promoted vigorous inflow and continuous convection. However, once Rolly made landfall over Catanduanes and Albay, frictional effects with rugged terrain, combined with increased vertical wind shear and dry air entrainment from the periphery of the subtropical ridge, rapidly disrupted the inner-core symmetry. The storm's central pressure rose sharply as its convection decoupled from the low-level circulation, initiating a rapid weakening trend. Within 24 hours after landfall, Rolly's eye had completely collapsed, its wind speeds had dropped by more than 100 km/h, and its circulation became disorganized as it traversed the West Philippine Sea, eventually dissipating into a tropical depression. This lifecycle encapsulates the delicate balance of environmental and internal factors governing tropical cyclone intensity—where even a perfectly structured system can quickly unravel once the environmental equilibrium is disturbed.

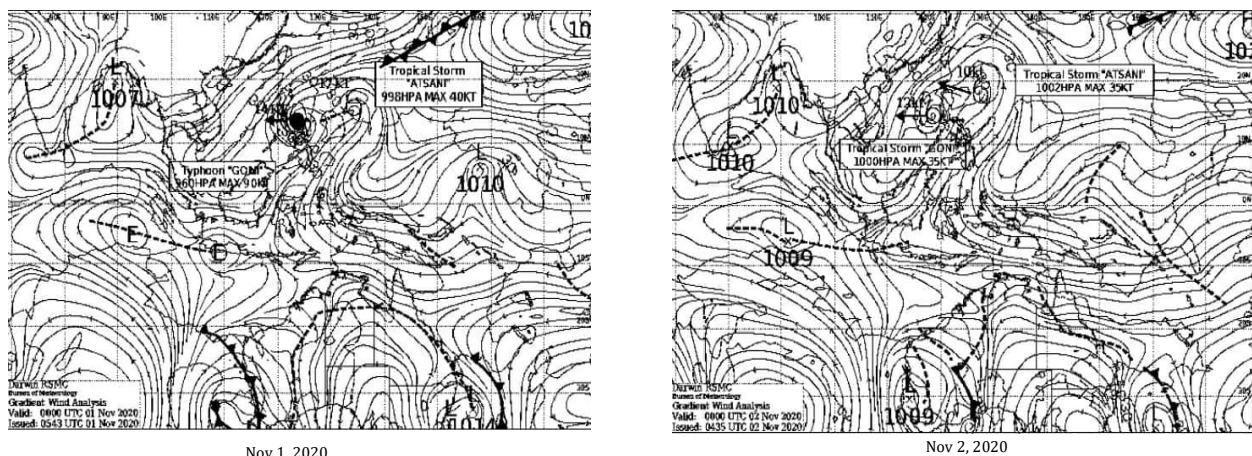


Figure 3. Wind field and pressure gradient analysis of Super Typhoon Rolly (JTWC, 2021).

V. Impacts and Hazards

Rolly's passage brought extreme rainfall, flooding, and destructive winds across the Bicol Region and Southern Luzon, leaving a trail of devastation that underscored the immense power of tropical cyclones interacting with complex terrain. According to PAGASA (2021), cumulative rainfall between October 29 and November 3 exceeded 350 mm in parts of Albay and Catanduanes, with the highest 24-hour totals surpassing 240 mm in Virac and Guinobatan. Orographic lifting significantly enhanced precipitation as warm, moisture-laden air was forced upward by the mountainous topography of Bicol, resulting in persistent and torrential rains (*Figure 4*). These intense downpours triggered widespread flooding and numerous landslides, while storm surges reaching up to 5 meters inundated coastal communities along Catanduanes and Albay's eastern seaboard. The heavy rainfall also remobilized volcanic ash and debris around Mayon Volcano, generating destructive lahar flows that engulfed several low-lying villages and damaged critical infrastructure. In addition to the hydrometeorological impacts, sustained winds exceeding 200 km/h flattened homes, uprooted trees, and toppled transmission lines, resulting in prolonged power outages and communication blackouts across multiple provinces. In total, Rolly caused 25 fatalities, 399 injuries, and damages estimated at ₱17.9 billion—₱12.9 billion in infrastructure and ₱5 billion in agriculture. Beyond the immediate destruction, secondary effects included large-scale displacement of families, contamination of water supplies, disruption of livelihoods, and long-term agricultural losses that strained food security in the months that followed, highlighting the compounding social and economic impacts of an extreme tropical cyclone on vulnerable communities.

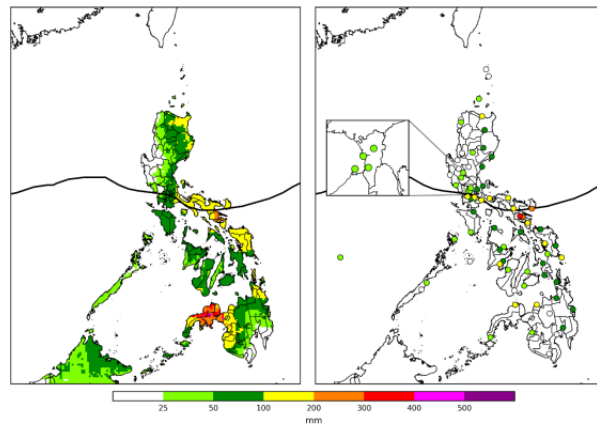


Figure 4. Rainfall totals recorded across Bicol Region from October 29 – November 3, 2020 (PAGASA, 2021).

VI. Concept Integration and Theoretical Application

Super Typhoon Rolly exemplifies numerous atmospheric processes covered in meteorology. Its formation relied on high sea surface temperatures that increased latent heat flux, while low wind shear allowed convection to organize around a low-pressure center. The system demonstrated hydrostatic instability and high Convective Available Potential Energy (CAPE), supporting deep convection. The Coriolis effect-initiated rotation, while the pressure gradient force drove violent winds around the core. Rolly's movement was influenced by the subtropical ridge—acting as a steering mechanism—and its outflow was enhanced by upper-tropospheric divergence. Figure 5 provides a conceptual diagram of tropical cyclone structure, highlighting its warm-core system, rainbands, and upper-level outflow patterns. These dynamics align with theories of tropical cyclone development discussed in class, particularly the Dvorak technique for estimating intensity.

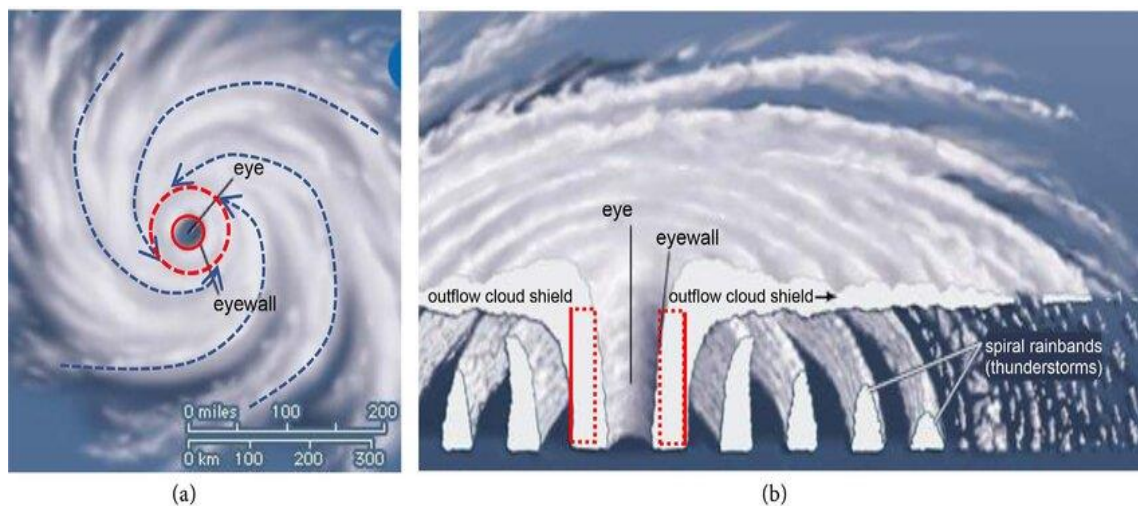


Figure 5. Conceptual diagram of a tropical cyclone structure showing eye, eyewall, and upper-level outflow (Qian W., 2020).

VII. Conclusion

Super Typhoon Rolly (Goni) stands as a historic and scientifically significant event. Its lifecycle demonstrates the interplay between oceanic heat, atmospheric instability, and upper-level dynamics in producing extreme weather phenomena. The storm underscored both the capabilities and limitations of current forecasting systems, as small-scale changes in the steering environment resulted in southward track adjustments. Rolly also highlighted the importance of translating scientific data into timely disaster response. Improved observation networks, real-time satellite analysis, and better communication between agencies remain critical for minimizing losses in future events. As future meteorologists, it is our role to integrate theoretical knowledge with practical forecasting and risk reduction efforts.

VIII. References

Asian Disaster Reduction Center (ADRC). (2020, November). Typhoon Goni (Rolly) situation report – Philippines (Situation Report No. 1). <https://www.adrc.asia>

CIMSS Satellite Blog. (2020, November 1). Typhoon Goni (Rolly) rapidly intensifies before landfall in the Philippines. Cooperative Institute for Meteorological Satellite Studies, University of Wisconsin–Madison. <https://cimss.ssec.wisc.edu/satellite-blog/archives/38784>

Earth Observatory (NASA). (2020, November 2). Devastating storm hits the Philippines (Typhoon Goni). <https://earthobservatory.nasa.gov/images/147476/devastating-storm-hits-the-philippines>

Japan Meteorological Agency (JMA). (2020). Best track data for Typhoon Goni (2020 #22). Regional Specialized Meteorological Center Tokyo–Typhoon Center. <https://www.jma.go.jp/jma/jma-eng/jma-center/rsmc-hp-pub-eg/besttrack.html>

Joint Typhoon Warning Center (JTWC). (2021). 2020 Annual Tropical Cyclone Report: Western North Pacific Basin. United States Navy. <https://www.metoc.navy.mil/jtwc/products/atcr/2020atcr.pdf>

PAGASA (DOST – Philippine Atmospheric, Geophysical and Astronomical Services Administration). (2021). Annual Report on Philippine Tropical Cyclones 2020 (ARTC 2020). https://pubfiles.pagasa.dost.gov.ph/pagasaweb/files/tamss/weather/tcsummary/PAGASA_ARTC_2020.pdf

UP Environmental Science Society. (2020, November 1). Advisory on Typhoon Goni (Rolly) and developing system Atsani (Siony) [Facebook post with wind-stream analysis]. Facebook. <https://www.facebook.com/upenvsci>